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RESEARCH INTERESTS

Measurement validity in model evaluation — whether the benchmarks and automatic judges used to rank models are measuring what they claim to. Concretely: counterfactual auditing of (multimodal) LLM judges, benchmark design grounded in large-scale real human behavior rather than model self-play, and social and strategic reasoning in LLM agents.

One result recurs across my projects: **surface competence $\neq$ underlying cognitive competence**. A correct answer is not a correct reasoning process; persuasive language is not strategic correctness; a verdict that looks right is not one that used the right evidence. I want to work on what is missing between generating language and thinking — measuring the gap with evaluation grounded in real human behavior, explaining it mechanistically, and closing it with interventions that change the underlying process rather than the benchmark number.

EDUCATION

Northeastern University, China — B.Eng., Computer Science and Technology 2023 – 2027 (expected)

- CGPA 4.05 / 5.00 (90 / 100) · Rank 5 / 100 in major cohort
- Research advising: Shiqi Zhao, Brain-Computer Interface and Hardware Acceleration Lab, since 2024

PUBLICATIONS

[1] Beyond Survival: Evaluating LLMs in Social Deduction Games with Human-Aligned Strategies

Zirui Song*, Yuan Huang*, Junchang Liu*, et al. (** equal contribution*)
EMNLP 2026, Main Conference · co-first author · arXiv:2510.11389 · Dataset

[2] Geolocation with Real Human Gameplay Data: A Large-Scale Dataset and Human-Like Reasoning Framework

Zirui Song, Jingpu Yang, Yuan Huang, et al.
Preprint · arXiv:2502.13759 · Dataset · Code

RESEARCH EXPERIENCE

EditJudgeBias — counterfactual auditing of multimodal image-editing judges

 2026 – present

Sole first author · conception, design, implementation, analysis, writing · in preparation, targeting ICLR 2027

Leaderboards for instruction-based image editing increasingly rank models with MLLM judges that are not themselves audited. This project makes "how much of the score is edit quality" measurable.

- **Method.** One invariant runs through the pipeline: admit only perturbations that can be shown not to have changed edit quality. Preservation verification is an admission gate every injector and metric must pass — not a post-hoc check — and each cue carries a zero-dose sham control establishing a per-judge, per-metric floor, so that null results are readable.
- **Scale.** 13 cue conditions across four injection sites — protocol, pixel, content, prompt — plus a null control; 5 published judges plus an independent replication arm with its own BH family; 3 disjoint content pools behind a near-duplicate gate. 97,973 judgments, 0.300% parse failure, human-annotation legs on 11 cues ($\kappa = 0.8930$, read beside $\kappa_{\max} = 0.8930$).
- **Findings.** Protocol-level reordering alone flips decisions at 38.2× the judge's self-noise. Rubric scoring buys absolute-score stability but not rank stability. Of three mitigation layers only the protocol-level one works, and its mechanism is **abstention, not debiasing**. A 5/5 null on a skin-tone attribute is readable only because same-batch positive controls fired — testing specificity alone would certify a never-alarming judge as best-calibrated (revocation sensitivity 0.032 vs 0.946 on identical data).
- **Verification.** 1,307 unit tests; 109 frozen artifacts each with sha256, producer, and an explicit "must not be read as" field; a self-certifying delivery package (1,289 tests pass in-package, 112 rebuilt tables byte-identical). The audit overturned four of my own conclusions, including a correction from 13/40 significant cells to 2/40 after adding clustering.

WereBench / WereAlign — human-aligned evaluation of LLM social reasoning

EMNLP 2026 Main

Co-first author · from project inception through submission

- Built a benchmark from 100+ hours of professionally produced human social-deduction gameplay (32.4M utterance tokens, 30 roles, 15 rule variants, multimodal), using winning-side MVP strategy trajectories as ground truth rather than LLM self-play.
- Designed a 5-dimension verbal evaluation and 2 decision tasks; contributed to research design, experiments, evaluation, and iterative refinement.
- Half of the evaluated models score below 0.50; the best reaches 0.720. Weakest dimensions are deception reasoning and counterfactual trade-off while persuasive generation stays strong — fluency ≠ strategic correctness. Two controlled interventions separated distinct failure paths, showing models can mistake another speaker's persuasive utterance for an instruction.

GeoComp / GeoCoT / GeoEval — geolocation from real human gameplay data

Preprint

Co-author (3 of 9)

- Contributed to a dataset from a public geo-guessing platform: 740K users, 2.7M locations, 25M human response records, with human accuracy calibrating item difficulty.
- Multi-step geographic reasoning improved standard metrics by up to 25%; scoring the reasoning process rather than only the final answer added a further 9%.

HONORS & AWARDS

- China National Scholarship — national level · 2023–2024
- Northeastern University Comprehensive Scholarship · 2024–2026
- Third Prize, National Finals, Lanqiao Cup Programming Competition · 2024

SKILLS

- Languages: Python, C++, Java · ML / research: PyTorch, Hugging Face Transformers
- English: IELTS Academic — taken 4 September 2026 (score to follow); CET-6 534
- Backend engineering intern, 2024 — Java / Spring Boot